

Problem Statement: Agentic AI System Using n8n for Automatic Technology Updates

Background

Technology evolves rapidly, making it difficult for students, developers, researchers, and professionals to stay updated with the latest trends, tools, frameworks, research papers, and industry developments. Manually tracking multiple sources is time-consuming and inefficient.

Problem Statement

Design and develop an Agentic AI system integrated with n8n that automatically gathers, filters, summarizes, and delivers relevant technology updates from multiple trusted sources. The system should act autonomously by monitoring predefined domains of interest and providing concise, personalized updates without requiring constant user intervention.

Objectives

- Automate the collection of technology-related news and updates.
- Integrate multiple data sources such as RSS feeds, blogs, newsletters, APIs, and research repositories.
- Use AI models to summarize and categorize collected information.
- Implement n8n workflows to orchestrate the end-to-end automation process.
- Deliver updates through channels such as email, messaging platforms, or dashboards.
- Reduce information overload by filtering irrelevant content.

Scope

- AI-driven content summarization and prioritization.
- Workflow automation using n8n.
- Scheduled execution and continuous monitoring.
- Personalized updates based on user interests.
- Multi-channel notification support.

Expected Outcome

The final system should function as an intelligent digital assistant capable of autonomously tracking emerging technologies, processing information, and delivering meaningful insights to users in an efficient and timely manner.

Technologies Used

- n8n for workflow automation
- Large Language Models (LLMs) for summarization

- RSS Feeds and APIs for data collection
- Email or messaging services for notifications
- Databases for storing historical updates

Potential Applications

The proposed solution can be used by educational institutions, software development teams, researchers, technology enthusiasts, and organizations seeking to remain informed about advancements in the technology landscape.